

INTERNAL COMBUSTION ENGINES

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DIVISION OF THERMAL AND ENERGY CONVERSION

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Introduction to IC Engine

CHAPTER ONE

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□ Contents

- Definition of Engine
- Definition of Heat Engine
- Heat Engine and classification
 - Internal combustion Engine
 - External combustion Engine
- Brief historical development of IC engines

Why we study about engine?

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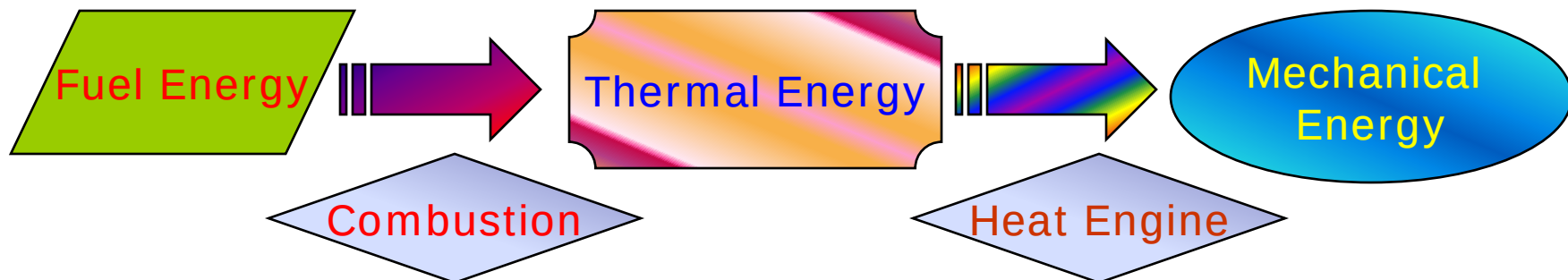
- Engines are one of the **greatest achievement** of the 20 century
- Engines are the foundation for the successful development of many important inventions
 - Automobile
 - Airplane
 - Agricultural mechanization
 - Petrochemical and mechanical technologies
- The internal-combustion engine will remain the "dominant" power source for vehicles until 2050

Definition of Engine

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□ Engine

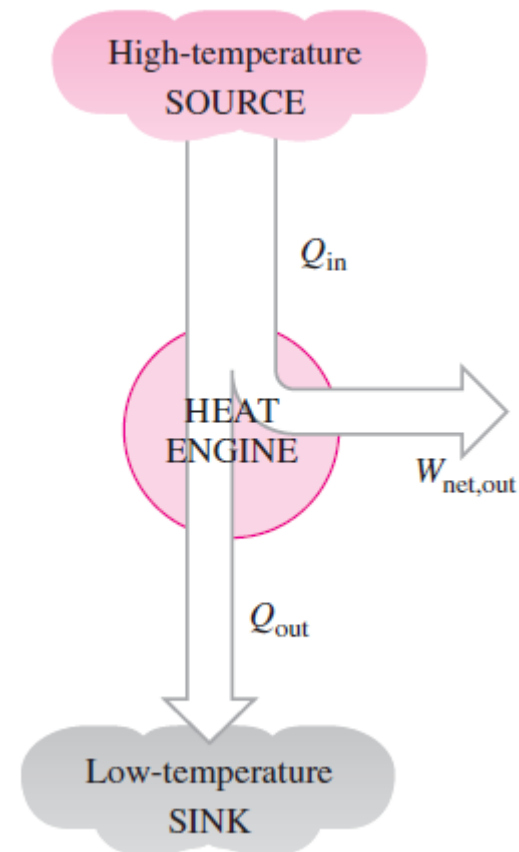
- is a device which transforms one form of energy in to another form.
- Most of the engines convert **Thermal Energy** into **Mechanical Work** and therefore they are called **Heat Engine**.



Heat Engines

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- Heat Engine is any device that is capable of converting **Thermal energy (heating)** into Mechanical energy (**work**).
- Heat engines differ considerably from one another, but all can be characterized by the following points :
 1. They receive heat from a **high-temperature source** (solar energy, oil furnace, nuclear reactor, etc.).
 2. They convert part of this heat to work (usually in the form of a rotating shaft).
 3. They reject the remaining waste heat to a **low-temperature sink** (the atmosphere, rivers, etc.).



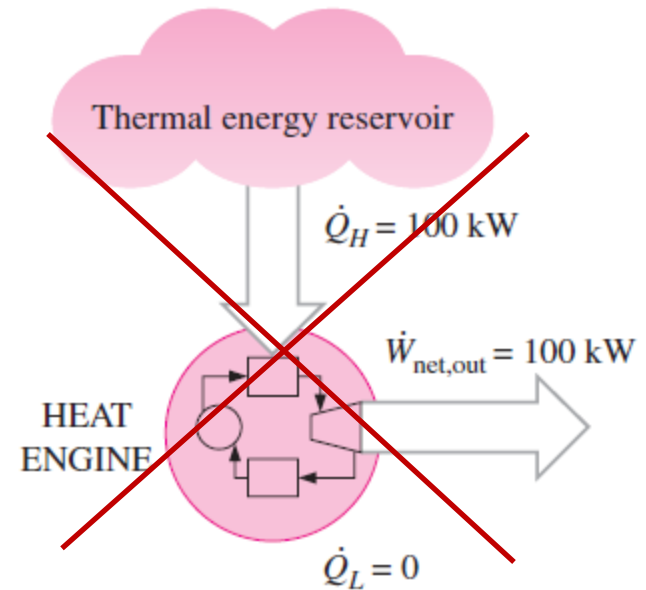
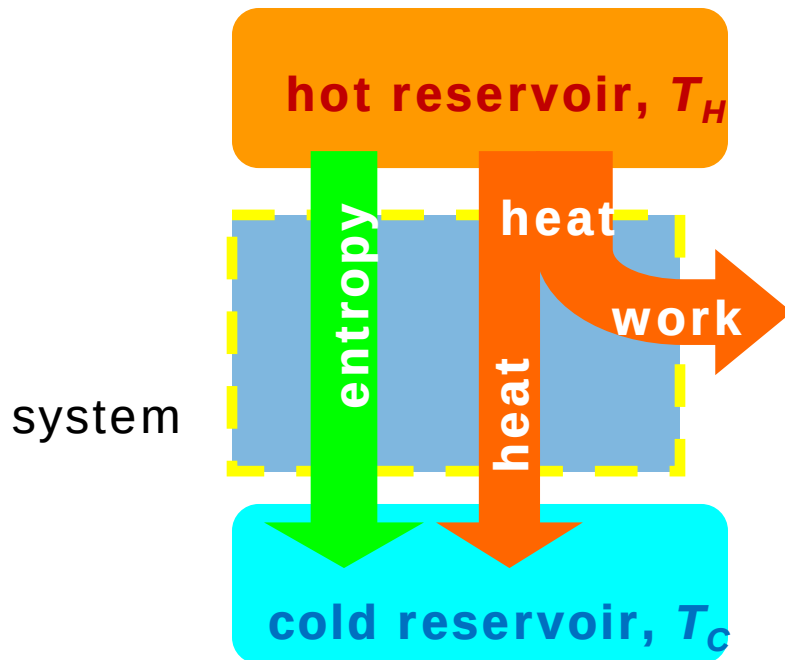
4. They operate on a cycle.

The Second Law of Thermodynamics

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□ Kelvin-Planck Statement

- It is impossible for any device that operates on a cycle to receive heat from a single reservoir and produce a net amount of work.



$$dS = \frac{\delta Q}{T}$$

Where:

dS - change in entropy

δQ - differential heat transfer

T -temperature at system boundary

Heat Engines

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- Heat engines usually involve a fluid to and from which heat is transferred while undergoing a cycle, called the working fluid.
- The term heat engine is often used in a broader sense to include work producing devices that do not operate in a thermodynamic cycle.
- Engines that involve internal combustion such as gas turbines and car engines fall into this category.
- IC engines operate in a mechanical cycle but not in a thermodynamic cycle since the working fluid (the combustion gases) does not undergo a complete cycle.
- Instead of being cooled to the initial temperature, the exhaust gases are purged and replaced by fresh air-and-fuel mixture at the end of the cycle.

Heat Engines

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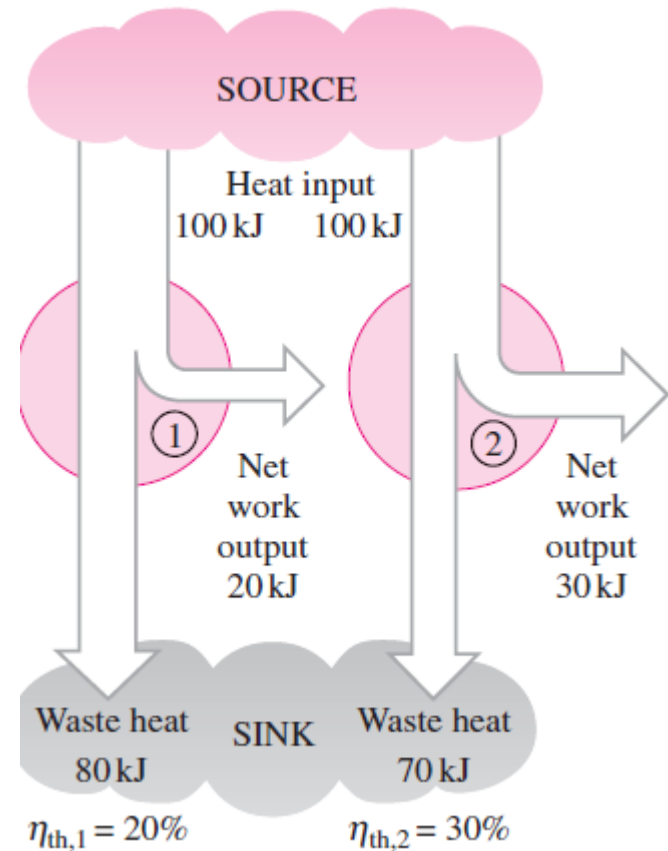
- Some heat engines perform better than others (convert more of the heat they receive to work).

$$\text{Thermal efficiency} = \frac{\text{Net work output}}{\text{Total heat input}}$$

$$\eta_{\text{th}} = \frac{W_{\text{net,out}}}{Q_{\text{in}}}$$

$$\eta_{\text{th}} = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$$

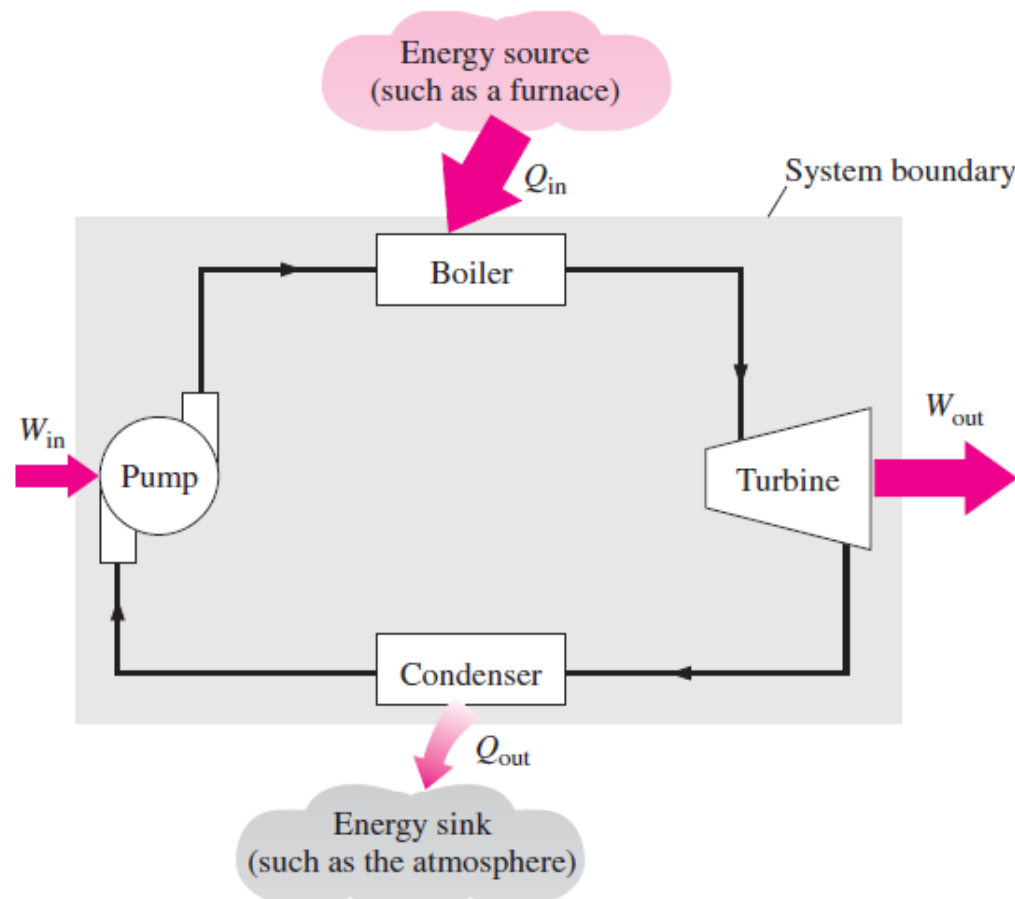
$$W_{\text{net,out}} = Q_{\text{in}} - Q_{\text{out}}$$



Heat Engines

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- The **work-producing device** that best fits into the definition of a **heat engine** is the **steam power plant**, which is an **external-combustion engine**.



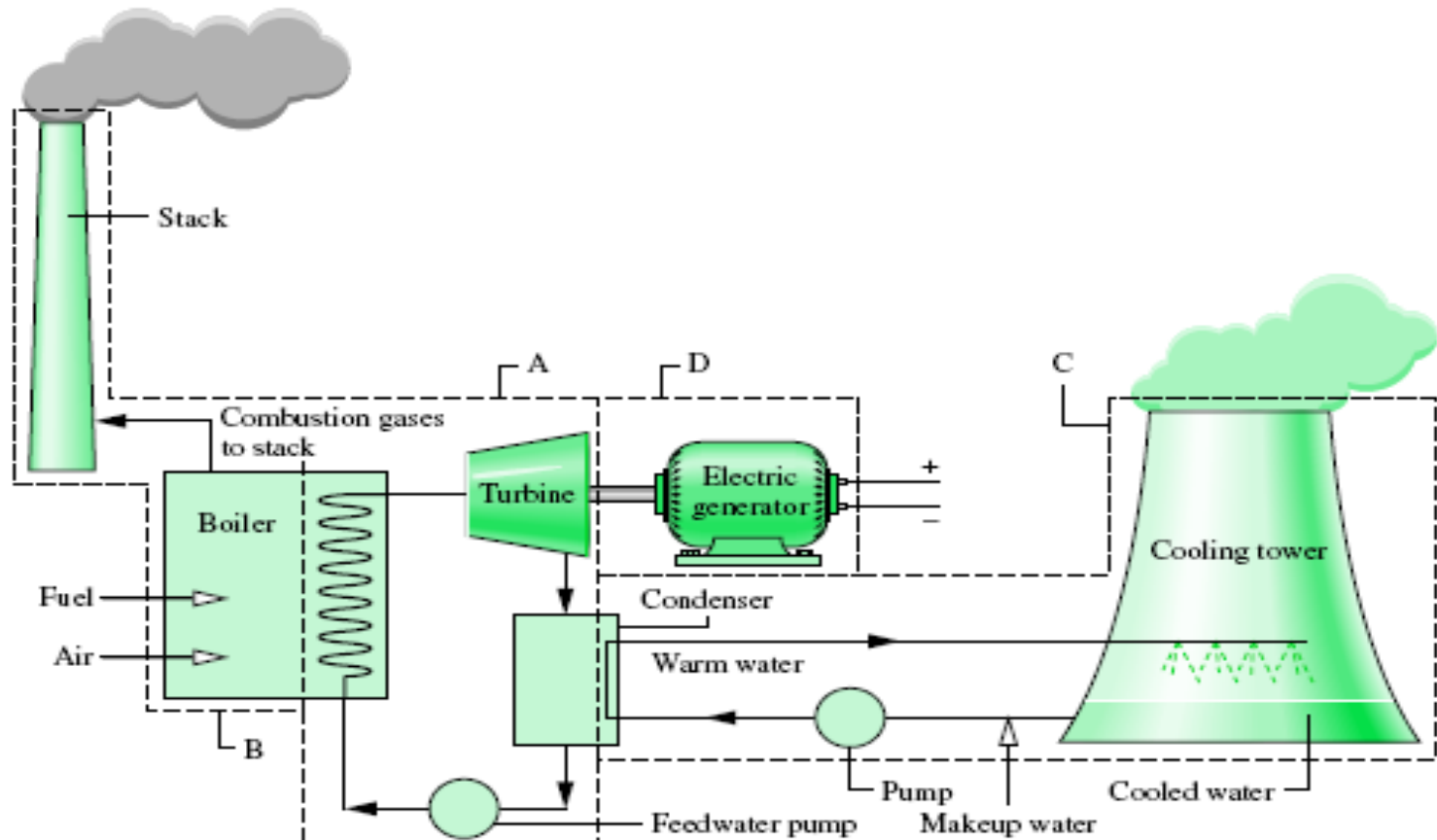
How EC differs from IC ?

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- ▶ The burning of fuel takes place outside the engine.
- ▶ A working fluid is utilized to transfer heat of combustion to the engine where heat is transformed into mechanical energy.
- ▶ A common example of this type is the steam power plant employing a boiler and a turbine
- ▶ Such an arrangement is not generally desirable for mobile power plants, since it entails the use of heavy and bulky heat exchangers, as well as the transportation of the supply working fluid.

Steam Power Plant (EC Engine)

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How IC differs from EC ?

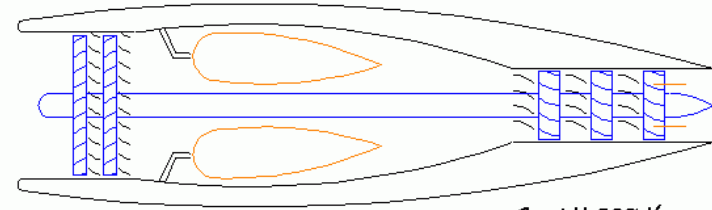
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- ▶ The combustion of a fossil fuel occurs in a combustion chamber in the portion of the engine which converts heat to mechanical energy.
- ▶ The expanding gases drive the engine directly
- ▶ The products of combustion are rejected back to the atmosphere.
- ▶ There is no necessity for an intermediate heat transferring apparatus, thus eliminating the need for heavy and bulky heat exchangers and the necessity of transporting the working fluid.

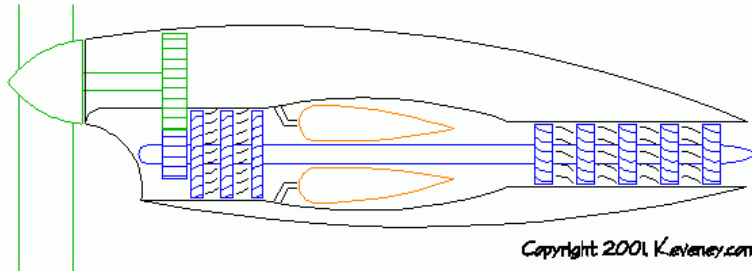
Gas Turbine Engines (IC Engines)

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Turbo-Jet



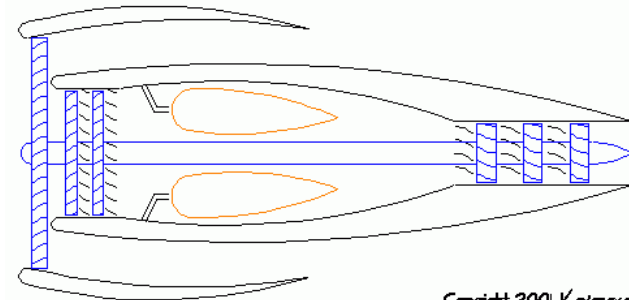
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Turbo-Prop

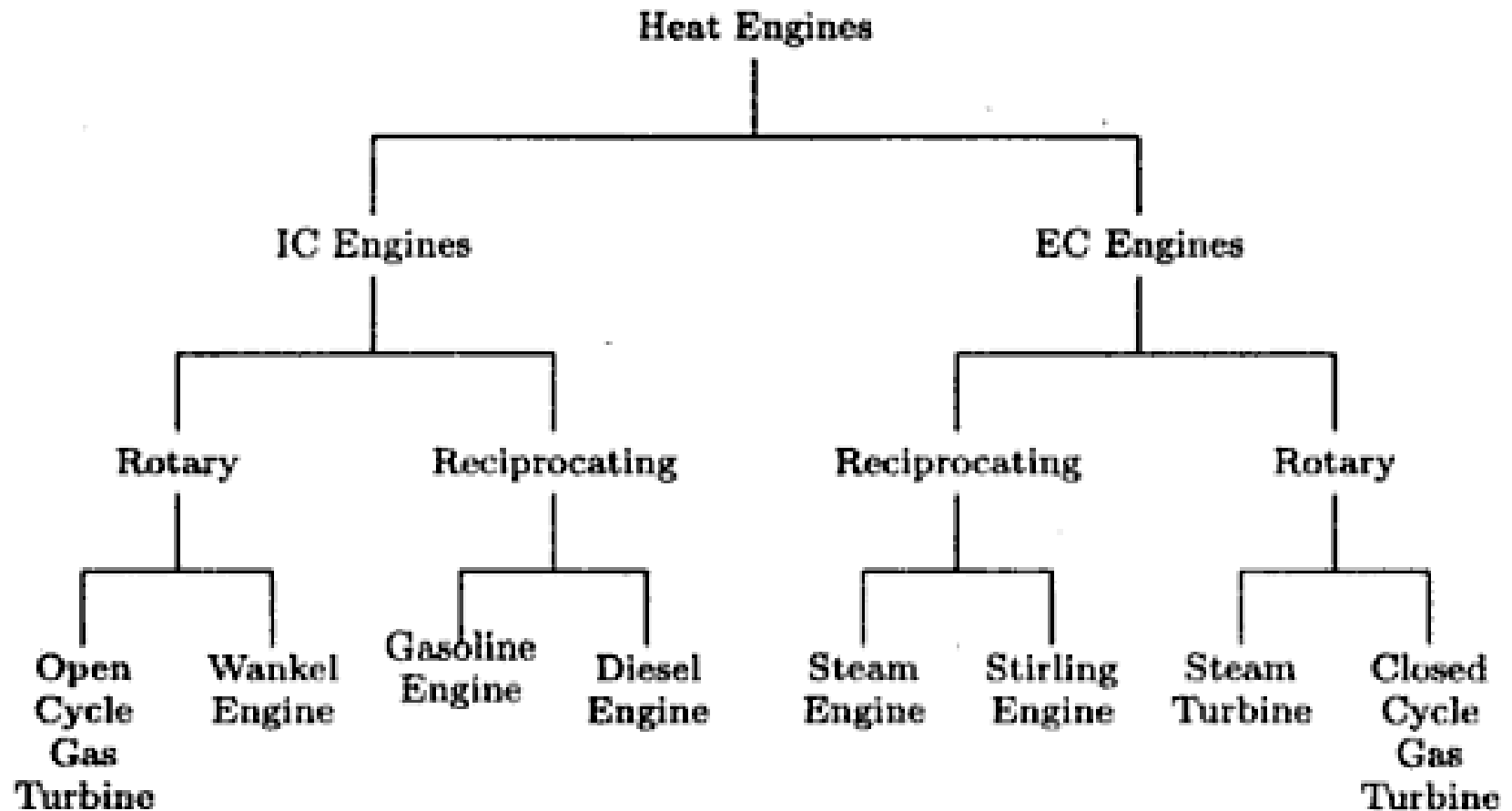
Turbo-Fan



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Summary of Heat Engines Classification

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Classification of Heat Engines

Internal Combustion Engines

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Internal Combustion Engine

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- The internal combustion engine (IC) is a heat engine that converts chemical energy in a fuel into Mechanical energy.
- Chemical energy of the fuel is first converted to thermal energy by means of combustion or oxidation with air inside the engine.
- This thermal energy raises the temperature and pressure of the gases within the engine, and the high-pressure gas then expands against the mechanical mechanisms of the engine.

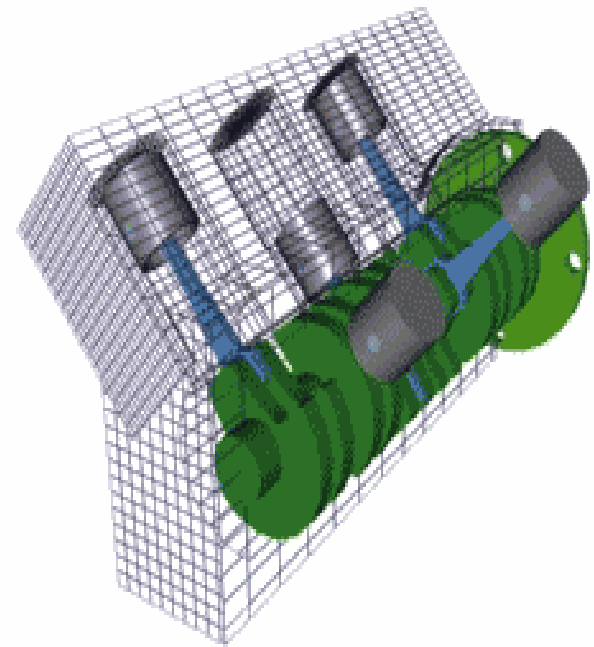
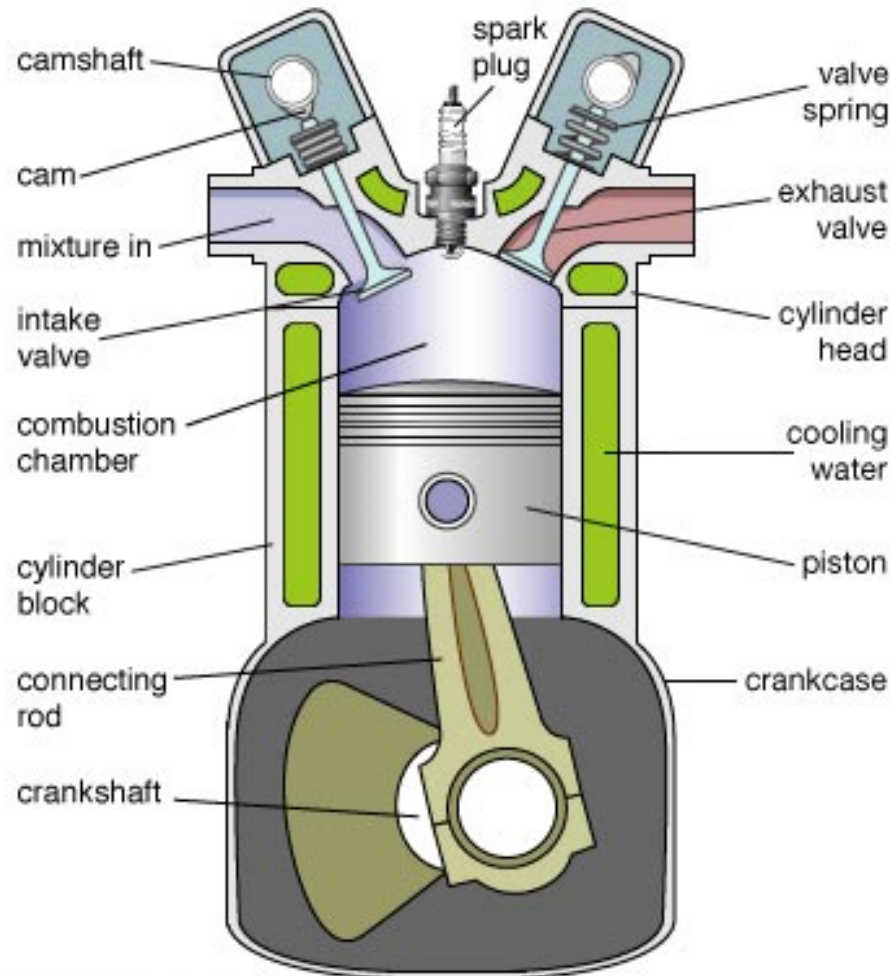
Internal Combustion Engine

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- This expansion is converted by the mechanical linkages of the engine to a rotating crankshaft, which is the output of the engine.
- The crankshaft, in turn, is connected to a transmission and/or power train to transmit the rotating mechanical energy to the desired final use.
- For engines this will often be the propulsion of a vehicle (i.e., automobile, truck, locomotive, marine vessel, or airplane).

Internal Combustion Engine

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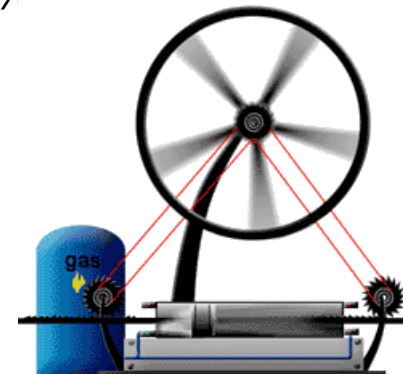
Historical Development of IC Engines

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- 1860: The first fairly practical engine was invented by J.J.E. Lenoir (1822-1900)
 - Single cylinder two stroke IC Engine
 - Burns a mixture of coal gas and air
 - Mixture was not compressed before ignition
- During the next decade, several hundred of these engines were built with power up to about 4.5 kW (6 hp) and mechanical efficiency up to 5%



Jean-Joseph-Etienne Lenoir
(1822 - 1900)



Historical Development of IC Engines

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❖ Otto-Langen Engine

- In 1867, the Otto-Langen engine, with efficiency improved to about 11%, was first introduced, and several thousand of these were produced during the next decade.
- This was a type of atmospheric engine with the power stroke propelled by atmospheric pressure acting against a vacuum.

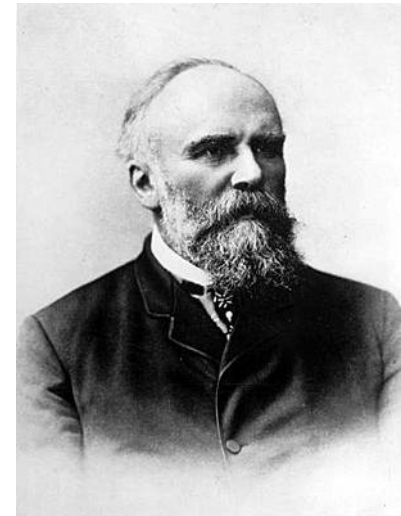
Historical Development of IC Engines

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❖ Otto- Langen Engine



Nicolaus A. Otto (1832-1891)



Eugen Langen (1833-1895)

Historical Development of IC Engines

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❖ Otto Engine

- In 1876, to overcome the shortcomings of low thermal efficiency and excessive weight, Otto proposed an engine cycle with four piston strokes:
 - ▣ an intake stroke,
 - ▣ a compression stroke before ignition,
 - ▣ an expansion or power stroke
 - ▣ an exhaust stroke.
- His prototype four-stroke engine first ran in 1876.

Historical Development of IC Engines

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❖ Karl Benz and Gottlieb Daimler gasoline engine

- In 1883, Karl Benz and Gottlieb Daimler, built an engine, where **gasoline** is induced into the induction air through a surface carburetor
- For the first time people succeeded in using **liquid fuels**.
- A large step was done toward the automobile with this, because liquid fuel needs less space than gaseous and can be transported more easily

Historical Development of IC Engines

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❖ Two-Stroke

- ▶ **1880**, Sir Dugald Clark developed the first **two-stroke internal combustion engines** where the **exhaust** and **intake processes** occur during the end of the **power stroke** and the beginning of the **compression stroke**.
- ▶ The **need to reduce or even replace the complicated valve mechanism** led to invention of the two stroke engine.
- ▶ The **need for less number of strokes than four strokes** and that performs **the same power as the Otto engine**

Historical Development of IC Engines

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❖ Diesel Engine

- 1892: The German engineer Rudolf Diesel (1858-1913) outlined in his patent a new form of internal combustion engine.
- His concept of initiating combustion by injecting a liquid fuel into air heated solely by compression permitted a doubling of efficiency over other internal combustion engines.
- Much greater expansion ratios, without detonation or knock, were now possible.

Historical Development of IC Engines

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❖ Wankel (Rotary Engine) (1929)

- Operate in four stroke principle

- **Advantages:**

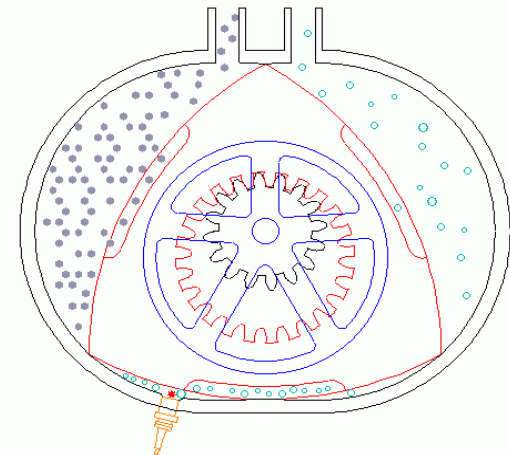
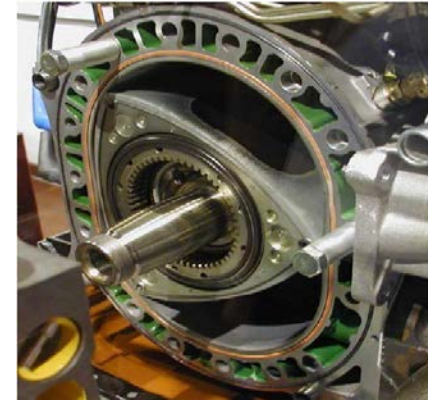
- High power output
- More reliable
- Simple structure and less moving parts
- Lower production cost
- Lighter and higher speed

- **Disadvantages :**

- Air-fuel mixing problem
- High hydrocarbon emissions
- Less fuel efficiency
- Sealing difficulty



Felix Wankel
(1902 - 1988)



Historical Development of IC Engines

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- Currently, five technologies that make IC engine better are
 - Clean diesel
 - Direct injection
 - Cylinder deactivation
 - Turbocharger
 - Variable valve timing
- The IC engine will remain the dominant power source for vehicles until 2050 if it is assisted by
 - Technology advancement
 - Infrastructure
 - Less manufacture cost

Historical Development of IC Engines

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- GM Chevrolet volt plug in Hybrid Electric Vehicle, Dec 2010
- With combined electric gasoline mode rated as 3.8l per 100km
- Electric motor is powered by batteries
- IC engine is powered by Gasoline or Diesel fuel
- PHEVs generally have larger packs than hybrid electric vehicles do
- PHEVs saves fuel, reduce emission, increase performance

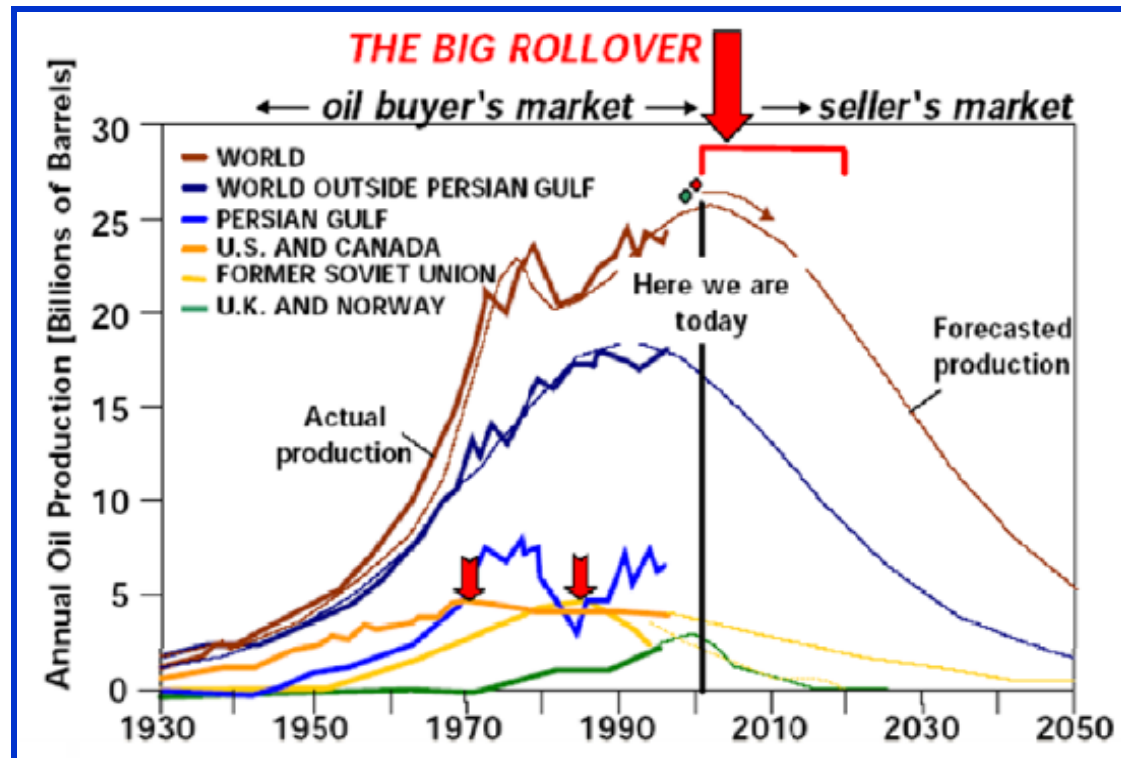


Best sold plug in electric hybrid vehicle in the world

Current Engine Challenges

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- ▣ Limited energy supply
- ▣ Global warming effect
- ▣ Environmental protection (Less pollutant emissions)

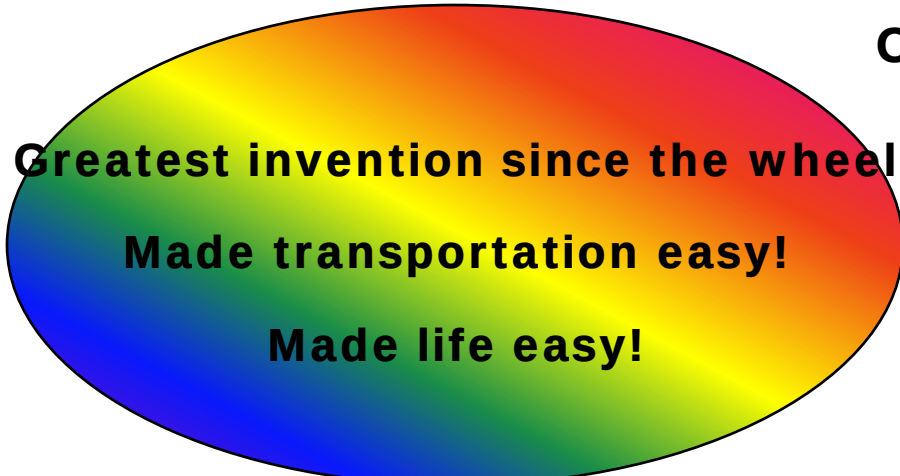


Why We Study about Engines?

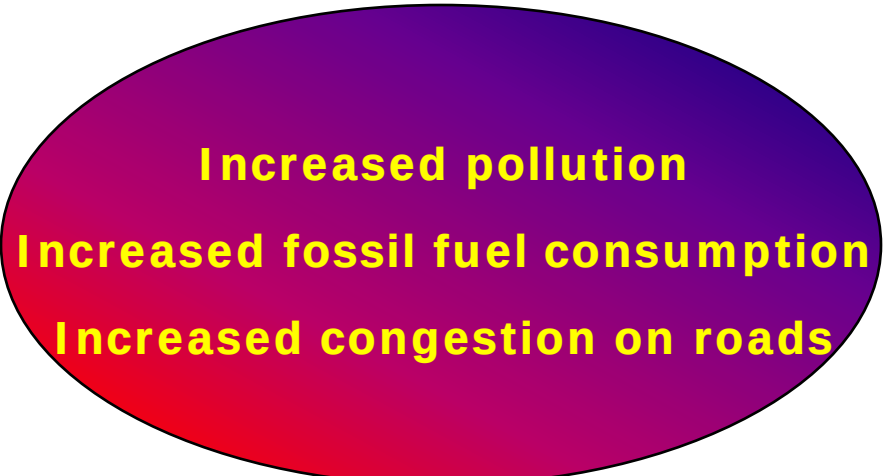
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ARE ENGINES **BOON** OR **BANE**?

OR



Greatest invention since the wheel
Made transportation easy!
Made life easy!



Increased pollution
Increased fossil fuel consumption
Increased congestion on roads

Why We Study about Engines?

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- Whether we like it or not....

CAN WE DO WITHOUT IT?

- Do we have viable alternatives?

THINK.....

- As of today we have no answer
- May be for at least 20 years more!

SO WE ARE STUCK WITH IT.....!

Classification of IC Engines

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- Engines can be classified according to the following criterias
 1. Application
 2. Basic Engine Design
 3. Operating Cycle
 4. Working Cycle
 5. Valve/Port Design and Location
 6. Fuel
 7. Mixture Preparation
 8. Ignition
 9. Stratification of Charge
 10. Combustion Chamber Design
 11. Method of Load Control
 12. Cooling

Classification of IC Engines

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1. APPLICATION

- Automotive
- Locomotive
- Light Aircraft
- Marine
- Power Generation
- Agricultural
- Earthmoving
- Home Use
- Others

Classification of IC Engines

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2. BASIC ENGINE DESIGN

I. Reciprocating

(a) Single Cylinder

(b) Multi-cylinder

- i. In-line
- ii. H, U,V,W & X
- iii. Radial
- iv. Opposed Cylinder
- v. Opposed Piston

II. Rotary

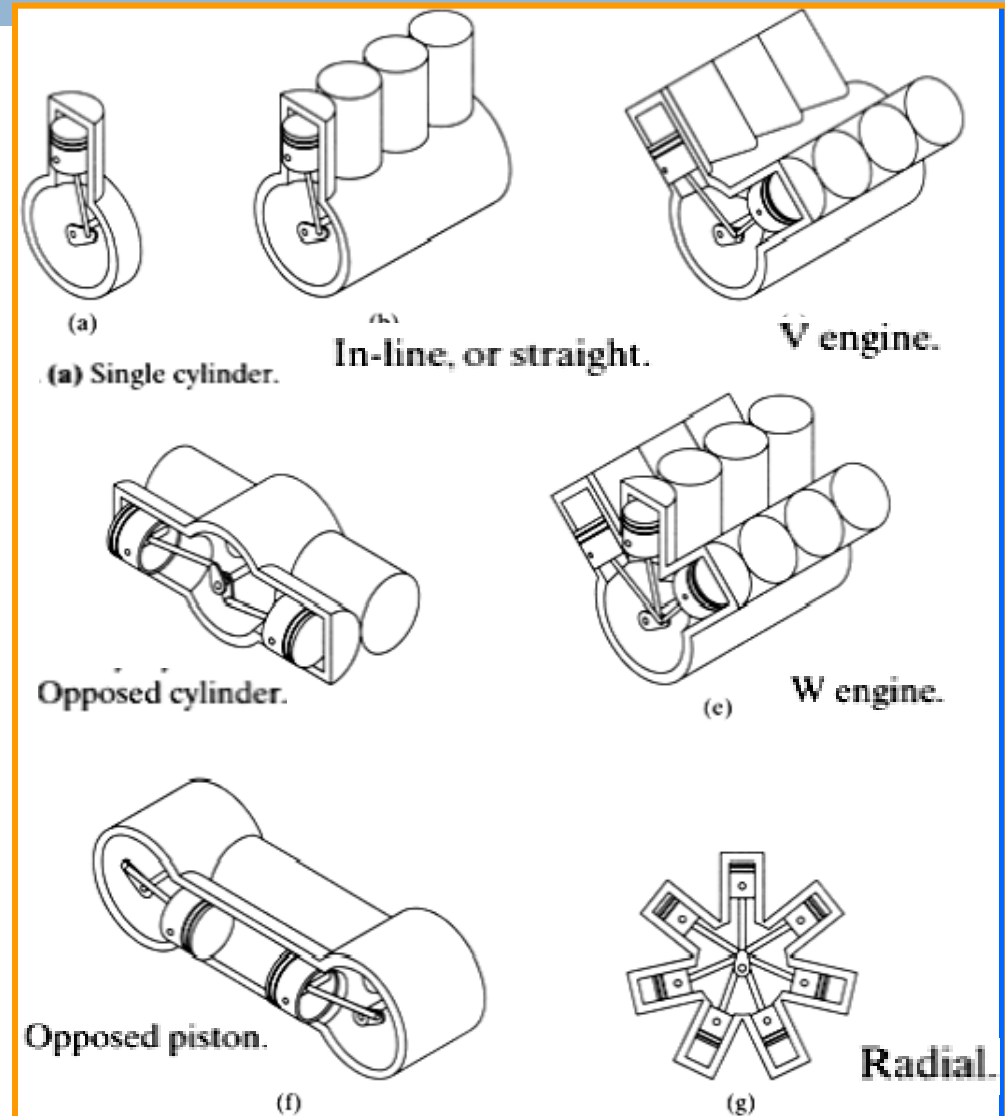
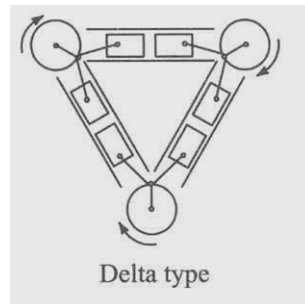
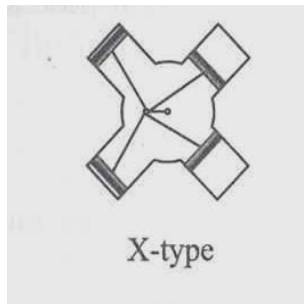
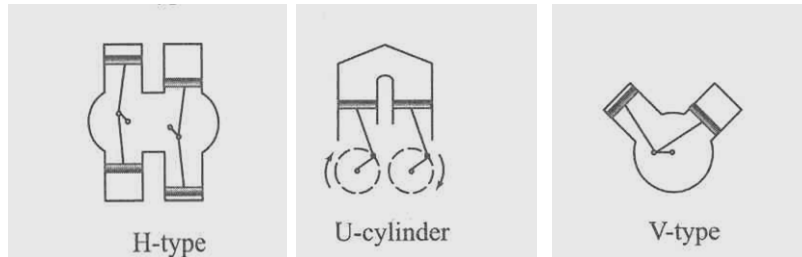
(a) Single Rotor

(b) Multi-rotor

Classification of IC Engines

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Position & Number of Cylinders



Classification of IC Engines

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a. Single Cylinder

- Engine has one cylinder and piston connected to the crankshaft.

b. In line

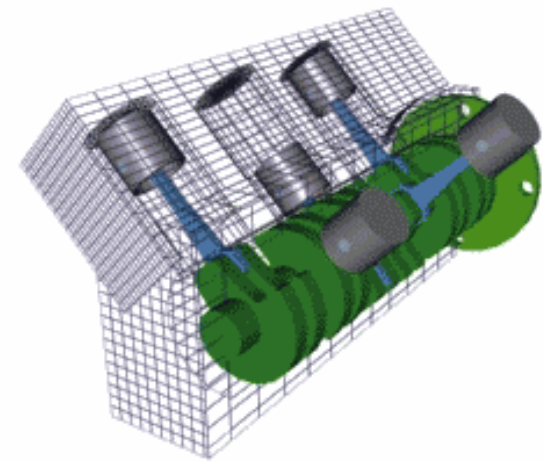
- Cylinders are positioned in a straight line, one behind the other along the length of the crankshaft. They can consist of 2 to 11 cylinders or possibly more.
- In-line four cylinder engines are very common for automobile and other applications. In-line engines are sometimes called straight. (e.g. Straight six or straight eight).

Classification of IC Engines

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c. V Engine

- Two banks of cylinders at an angle with each other along a single crankshaft.
- The angle between the banks of cylinders can be anywhere from 15° to 120° , with 60° - 90° being common. V engines have even numbers of cylinders from 2 to 20 or more.

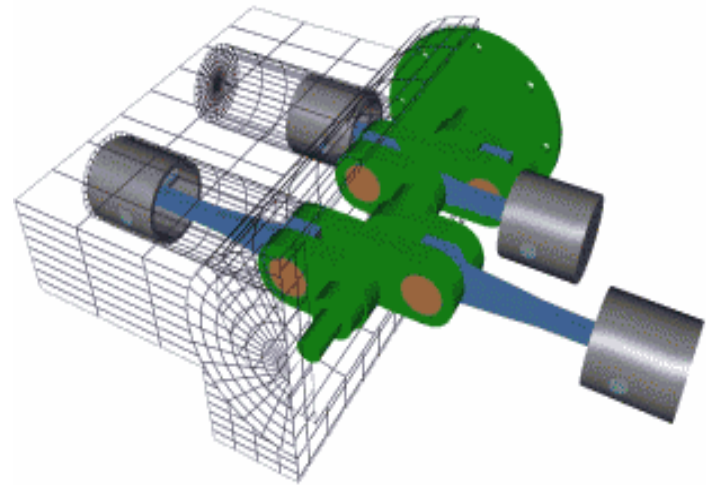


Classification of IC Engines

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d. Opposed Cylinder Engine

- Two banks of cylinders opposite each other on a single crankshaft (a V engine with a 180° V).
- These are common on small aircraft and some automobiles with an even number of cylinders from two to eight or more. These engines are often called flat engines (e.g., flat four).

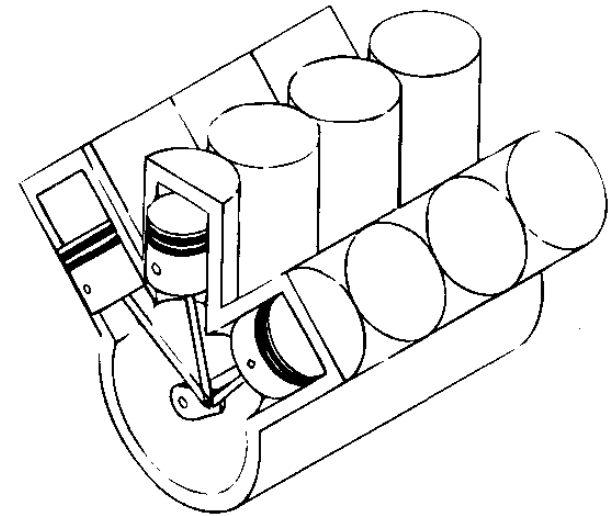


Classification of IC Engines

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e. W Engine

- Same as a V engine except with three banks of cylinders on the same crankshaft.
- Not common, but some have been developed for racing automobiles, both modern and historic.
- Usually 12 cylinders with about a 60° angle between each bank.

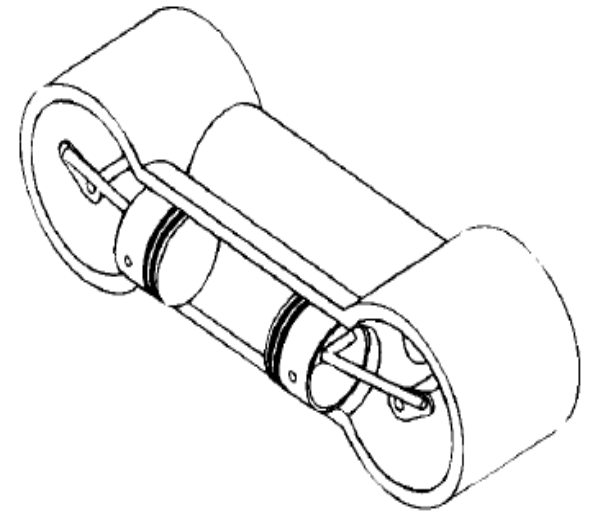


Classification of IC Engines

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f. Opposed Piston Engine

- Two pistons in each cylinder with the combustion chamber in the center between the pistons.
- A single-combustion process causes two power strokes at the same time, with each piston being pushed away from the center and delivering power to a separate crankshaft at each end of the cylinder.
- Engine output is either on two rotating crankshafts or on one crankshaft incorporating complex mechanical linkage.



Classification of IC Engines

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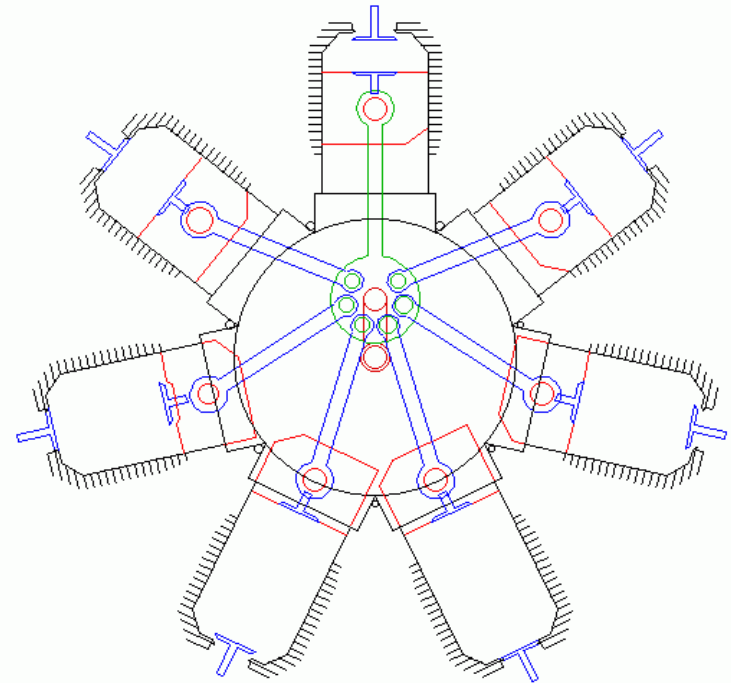
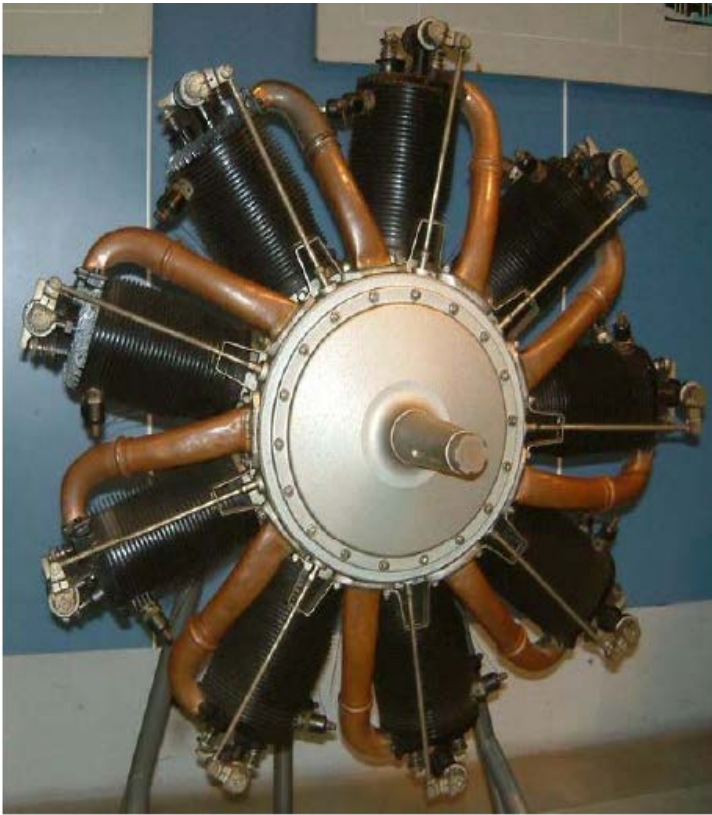
g. Radial Engine

- Engine with pistons positioned in a circular plane around the central crankshaft.
- The connecting rods of the pistons are connected to a master rod which, in turn, is connected to the crankshaft.
- A bank of cylinders on a radial engine always has an odd number of cylinders ranging from 3 to 13 or more.

Classification of IC Engines

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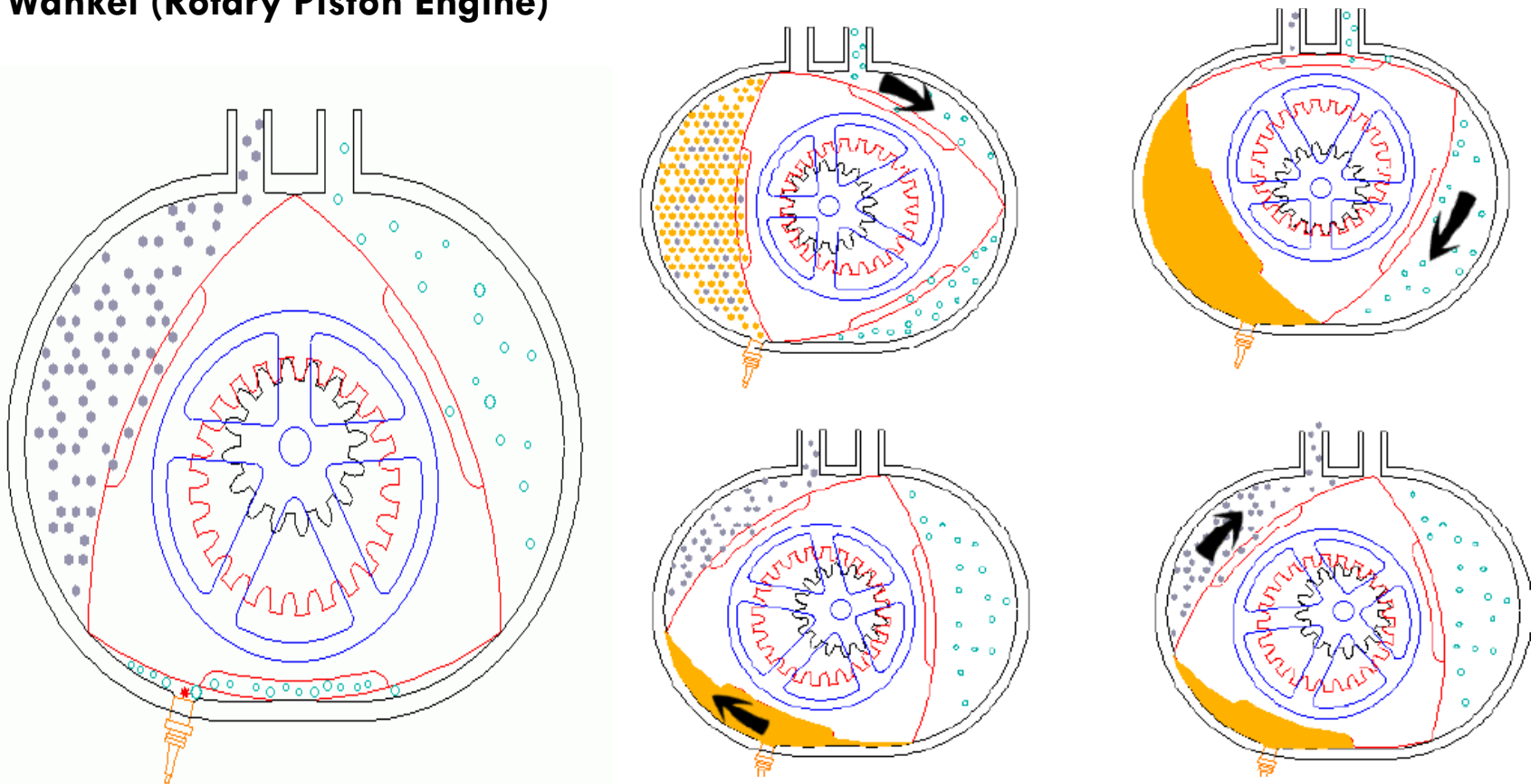
□ Radial Engine



Classification of IC Engines

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Wankel (Rotary Piston Engine)



Classification of IC Engines

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3. OPERATING CYCLE

- Otto (For the Conventional SI Engine)
- Atkinson (For Complete Expansion SI Engine)
- Miller (For Early or Late Inlet Valve Closing type SI Engine)
- Diesel (For the Ideal Diesel Engine)
- Dual (For the Actual Diesel Engine)

Classification of IC Engines

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4. METHOD OF INCREASING INLET PRESSURE (POWER BOOSTING)

1. Naturally Aspired

- No intake air pressure boost system

2. Supercharger

- Intake air pressure increased with the compressor driven off of the engine crankshaft.

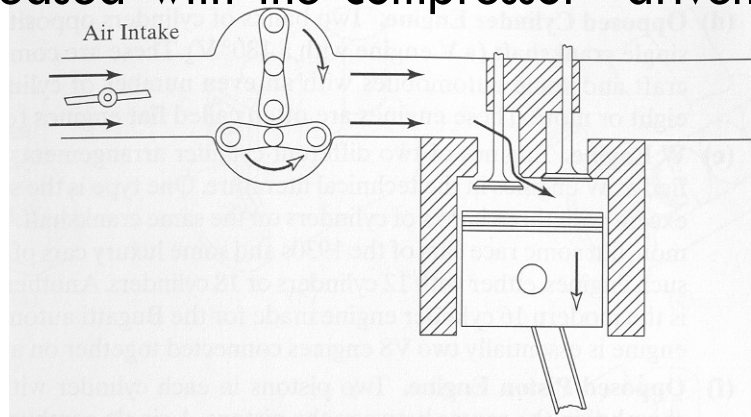


FIGURE 1-8

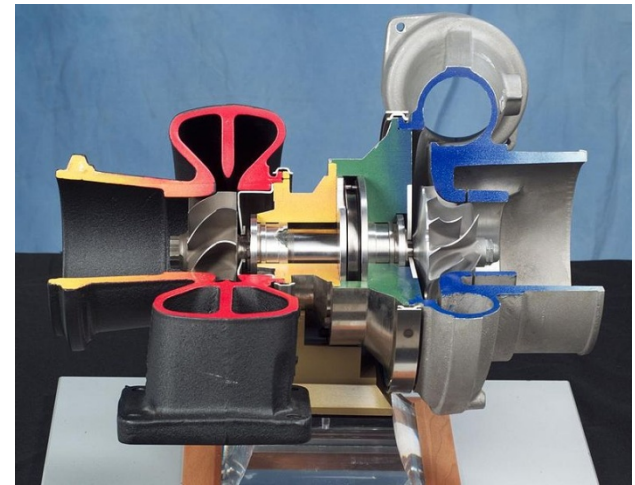
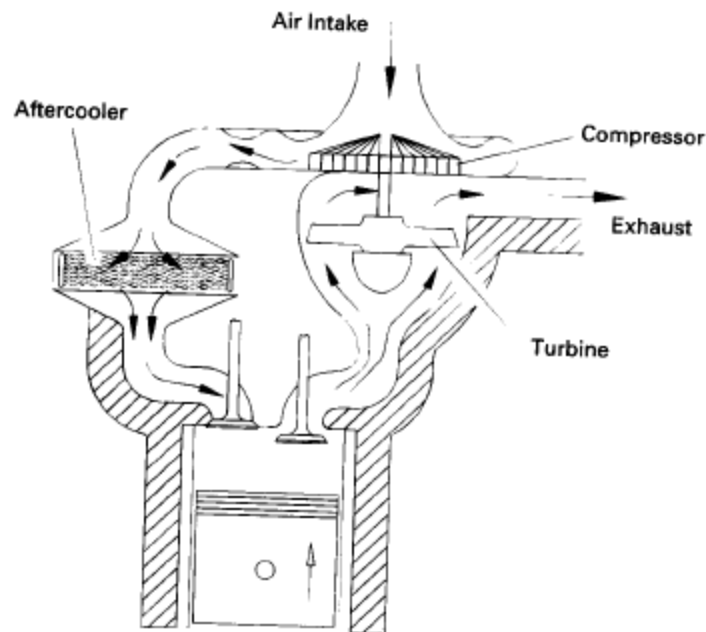
Supercharger used to increase inlet air pressure to engine. Compressor is driven off engine crankshaft, which gives fast response to speed changes but adds parasitic load to engine.

Classification of IC Engines

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3. Turbocharged

- Intake air pressure increased with the turbine-compressor driven by the engine exhaust gases.

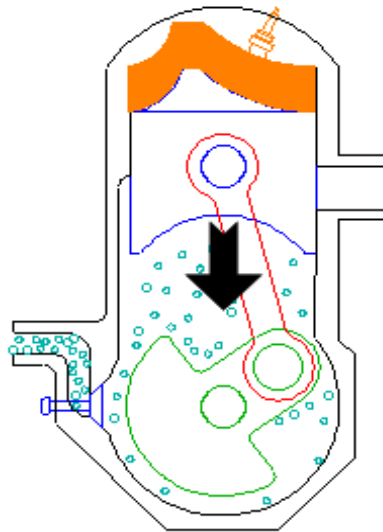


Classification of IC Engines

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4. Crankcase Compressed

- Two Stroke cycle engine which uses the crankcase as the intake air compressor.
- Limited development work has also been done on design and construction of four stroke cycle engines with crankcase compression.



Classification of IC Engines

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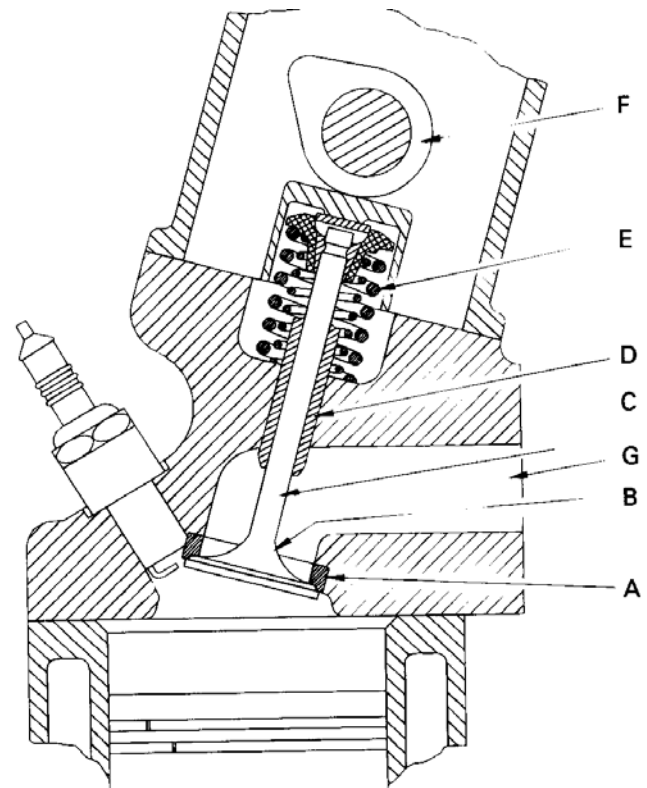
5. VALVE/PORT DESIGN AND LOCATION

Design

1. Poppet Valve
2. Rotary Valve
3. Reed Valve
4. Piston Controlled Porting

Location

1. The T-head
2. The L-head
3. The F-head
4. The I-head:
 - (i) Over head Valve (OHV)
 - (ii) Over head Cam (OHC)



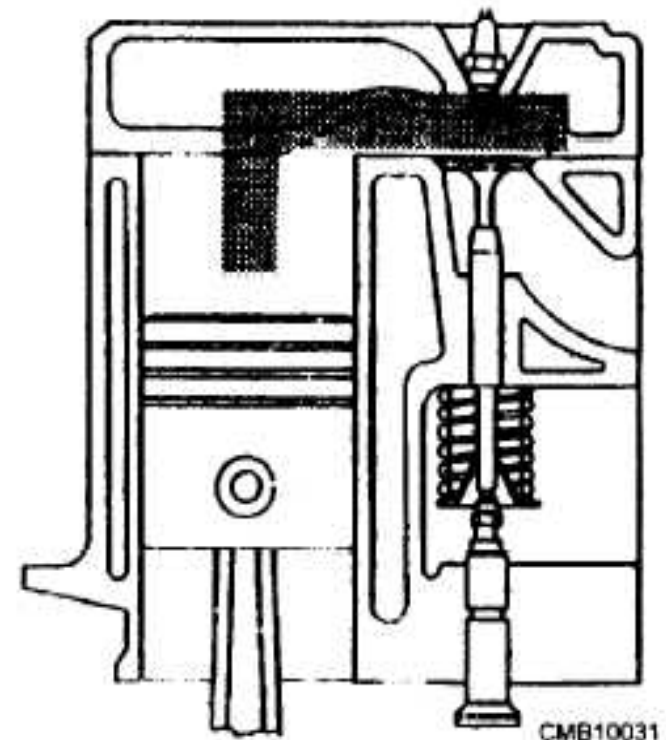
Classification of IC Engines

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- According to the arrangement of the intake and exhaust valves, whether the valves are located in the cylinder head or cylinder block.

L-HEAD

- The intake and the exhaust valves are both located on the same side of the piston and cylinder.
- The valve operating mechanism is located directly below the valves, and one camshaft actuates both the intake and the exhaust valves

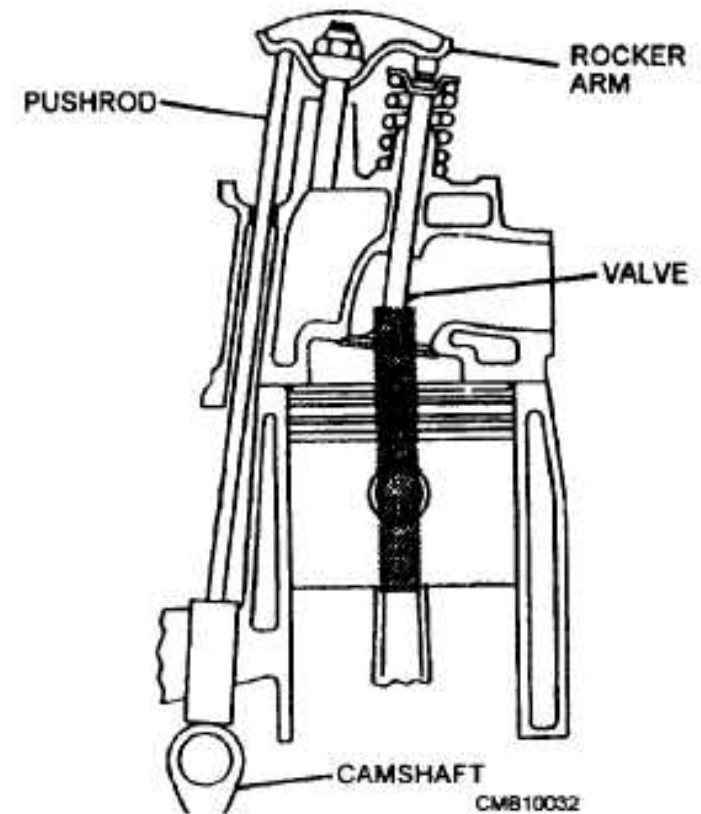


Classification of IC Engines

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I-HEAD

- The intake and the exhaust valves are both mounted in a cylinder head directly above the cylinder.
- This arrangement requires a tappet, a pushrod, and a rocker arm above the cylinder to reverse the direction of valve movement.
- Although this configuration is the most popular for current gasoline and diesel engines.
- It was rapidly superseded by the overhead camshaft.

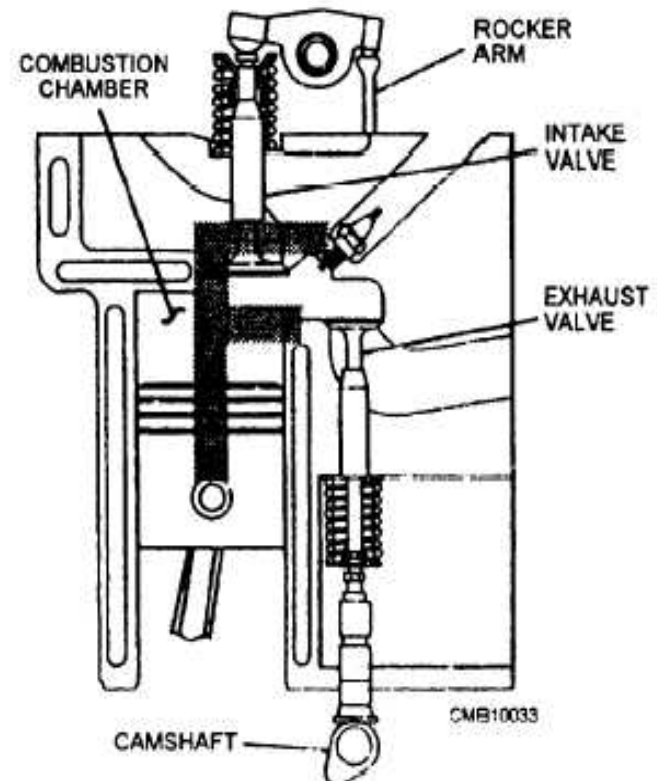


Classification of IC Engines

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F-HEAD

- The intake valves are normally located in the head, while the exhaust valves are located in the engine block.
- The intake valves in the head are actuated from the camshaft through tappets, pushrods, and rocker arms.
- The exhaust valves are actuated directly by tappets on the camshaft.

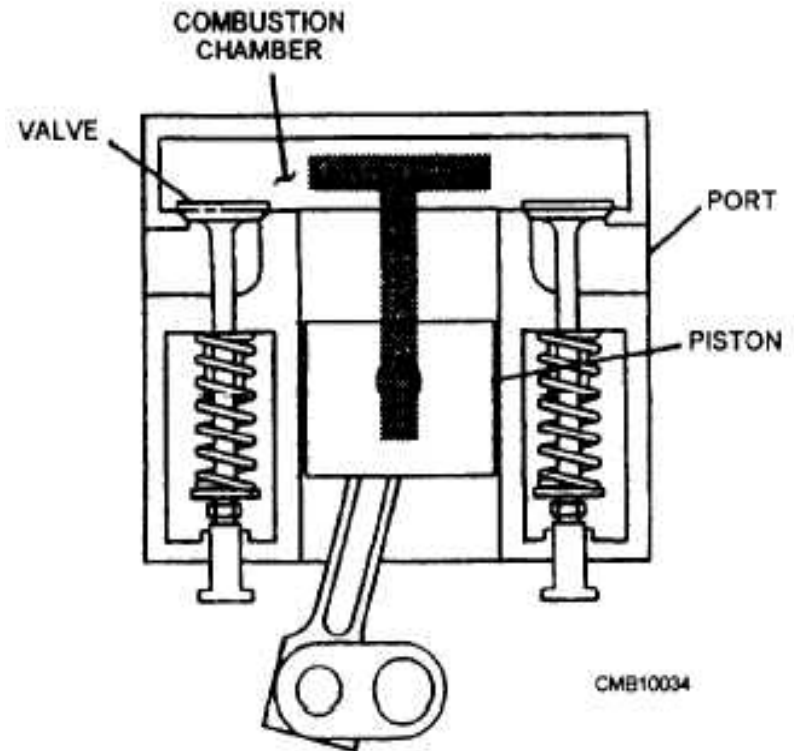


Classification of IC Engines

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T-HEAD-

➤ The intake and the exhaust valves are located on opposite sides of the cylinder in the engine block, each requires their own camshaft.



Classification of IC Engines

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6. FUEL

1. Conventional

(a) Crude oil derived

- (i) Petrol
- (ii) Diesel

2. Alternate

- (b) Bio-mass Derived
- (i) Alcohols (methyl and ethyl)
 - (ii) Vegetable oils
 - (iii) Producer gas and biogas
 - (iv) Hydrogen

3. Blending

4. Dual fueling

Classification of IC Engines

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7. MIXTURE PREPARATION

1. Carburetion

2. Fuel Injection

- (i) Diesel

- (ii) Gasoline

- (a) Port

- (b) Cylinder

Classification of IC Engines

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8. BASED ON TYPE OF IGNITION

1. Spark Ignition (SI)

- The engine starts the combustion process in each cycle by use of a spark plug.

2. Compression Ignition (CI)

- The combustion process in a CI engine starts when the air-fuel mixture self-ignites due to high temperature in the combustion chamber caused by high compression.

Classification of IC Engines

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9. BASED ON ENGINE CYCLE

1. Four-Stroke Cycle

- A four-stroke cycle experiences four piston movements over two engine revolutions of each cycle

2. Two-Stroke Cycle

- A two-stroke cycle has two piston movements over one revolution for each cycle

Classification of IC Engines

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10. METHOD OF LOAD CONTROL

1. **Throttling**

- To keep mixture strength constant
- Also called Charge Control Used in the Carbureted SI Engine

2. **Fuel Control**

- To vary the mixture strength according to load, used in the CI Engine

3. **Combination**

- Used in the Fuel-injected SI Engine.

Classification of IC Engines

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11. COOLING

1. Direct Air-cooling
2. Indirect Air-cooling (Liquid Cooling)
3. Low Heat Rejection (Semi-adiabatic) engine.

Basic Engine Componentes

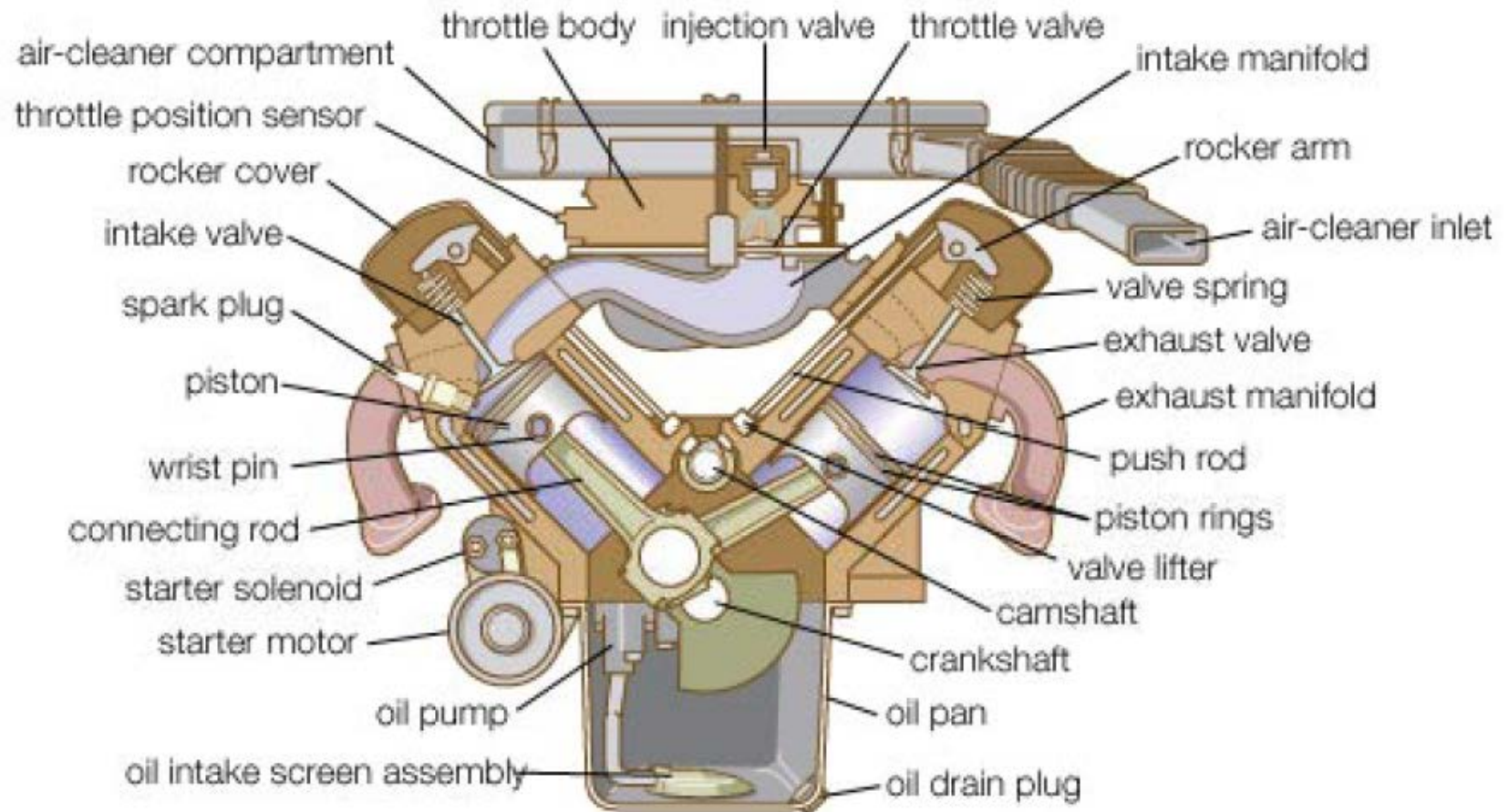
59

❖ Major Engine Parts

- ❑ Cylinder Block
- ❑ Cylinder Head
- ❑ Crankshaft
- ❑ Camshaft
- ❑ Timing Chain
- ❑ Bearing shell
- ❑ Oil pump
- ❑ Water pump
- ❑ Fly wheel
- ❑ Valves
- ❑ Valve Springs
- ❑ Pistons
- ❑ Connecting Rod
- ❑ Piston Ring
- ❑ Cylinder sleeve
- ❑ Inlet manifold
- ❑ Exhaust manifold
- ❑ Rocker Arm

Basic Engine Components

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IC Engine construction

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- ❖ Engine construction can be broken down into two categories:- Stationary parts and Moving parts

- ❖ Stationary parts

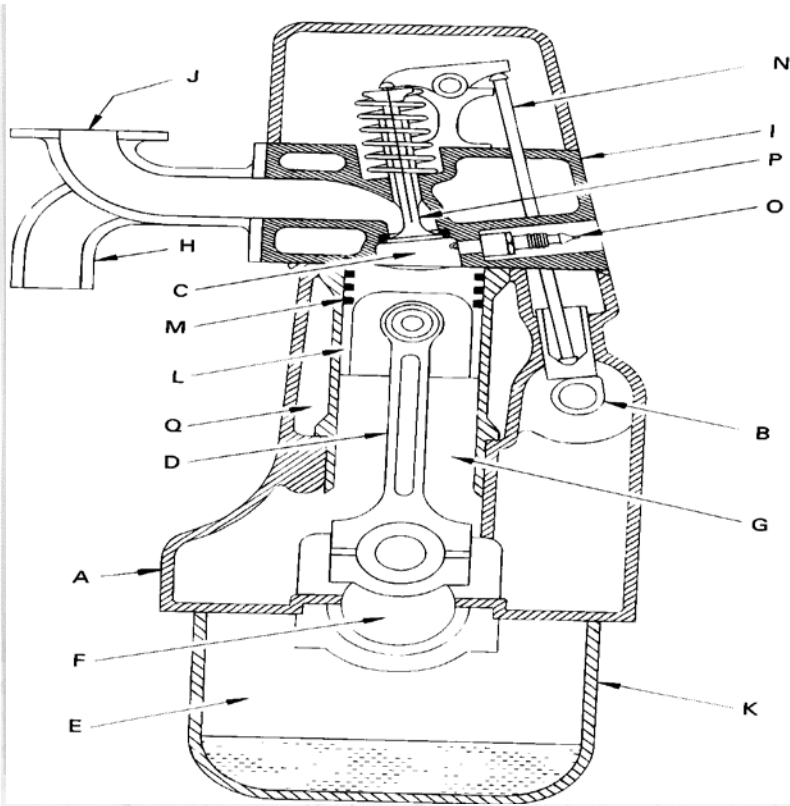
The stationary parts of an engine include

- Cylinder block,
 - Cylinders,
 - Cylinder head or heads,
 - Crankcase, and the exhaust and intake manifolds.
- These parts furnish the framework of the engine.
- All movable parts are attached to or fitted into this framework.

Stationary Part

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❖ Cylinder Block



“Backbone” of the engine.
Supports / aligns most
other components.

Contains:

- Cylinders
- Coolant passages
- Oil passages
- Bearings

One-piece, gray cast iron

Stationary Part

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❖ Cylinders

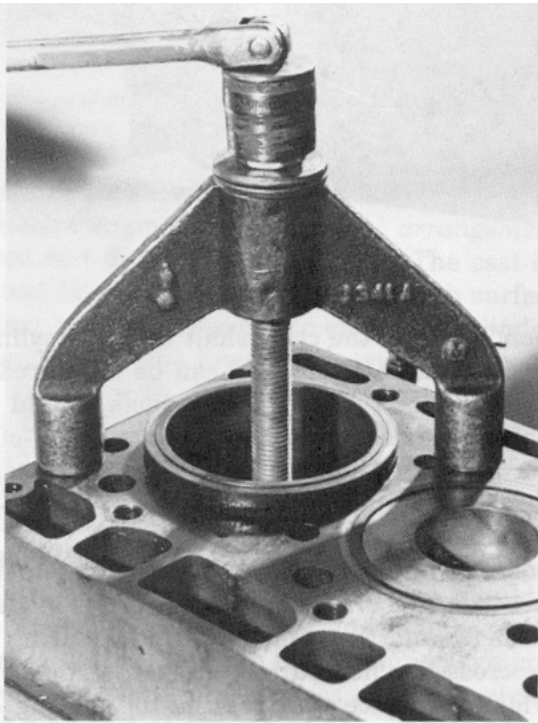


Figure 3.4—Removal of a cylinder liner. (Photo by Laurie Goering.)

- Cylindrical holes in which the pistons reciprocate.
- May be:
 - Enblock
 - Liners
 - Wet liners
 - Dry liners
- Cylinder bore – diameter of cylinder

Stationary Part

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❖ Cylinder Head

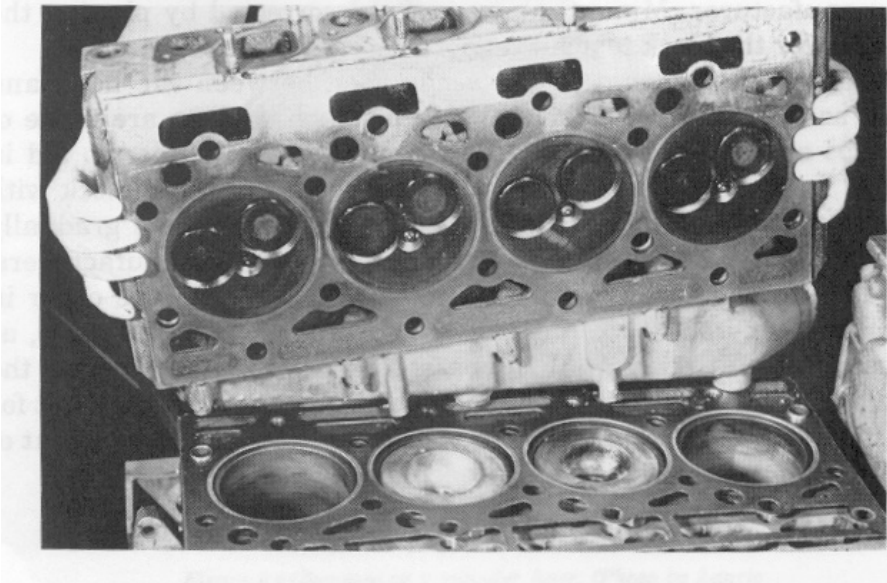
Seals the “top-end” of the combustion chamber.

Head bolts and head gasket ensure air-tight seal of the combustion chamber.

Contains the valves and the intake and exhaust “ports”.

Contains oil and coolant passages.

One –piece castings of iron alloy



Stationary Part

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❖ Crankcase

- ▶ The crankcase is that part of the engine block below the cylinders.
- ▶ It supports and encloses the crankshaft and provides a reservoir for the lubricating oil
- ▶ Contains a place for mounting the oil pump, oil filter, starting motor
- ▶ The lower part of the crankcase is the OIL PAN, which is bolted at the bottom. Is used as a reservoir for collecting and holding lube oil

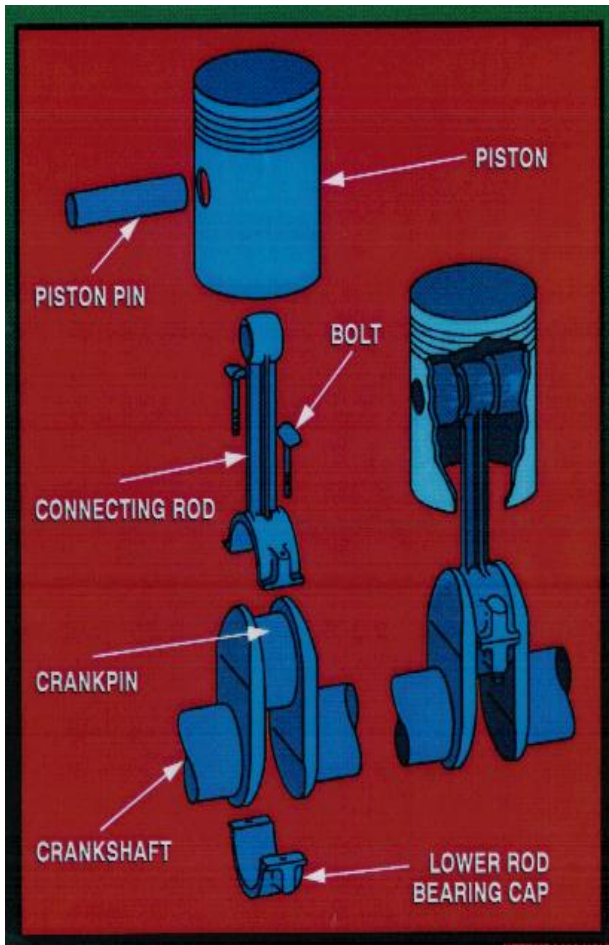
Moving Components

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- Moving parts contains three groups according to their motion
 - Reciprocating only (pistons and valves)
 - Reciprocation & rotary (connecting rods)
 - Rotary only (crankshafts and camshafts)

Moving Components

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❖ Piston

- ❑ Forms the “moveable bottom” of the combustion chamber.
- ❑ Lightweight but strong/durable

❑ Piston Rings

➤ Oil ring and air ring

- ❑ Transfer heat from piston to cylinder
- ❑ Seal cylinder & distribute lube oil

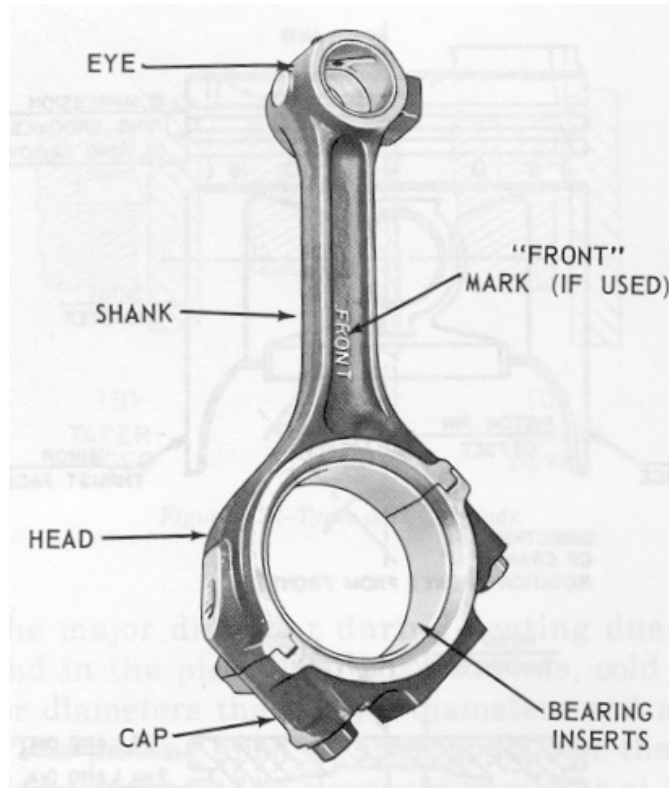
❑ Piston Pin

- ❑ Pivot point connecting piston to connecting rod

Moving Components

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❖ Connecting Rod

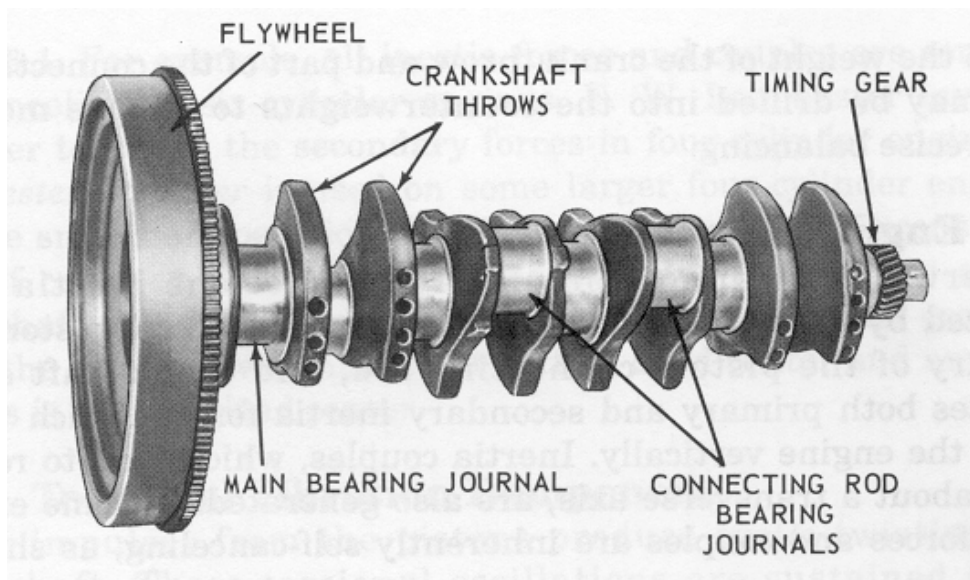


- Connects the piston to the crankshaft
- Converts reciprocating piston motion to rotary motion at the crankshaft.
- Drop-forged steel

Moving Components

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❖ Crankshaft



➤ Works with connecting rod to change reciprocating motion of the piston to rotary motion

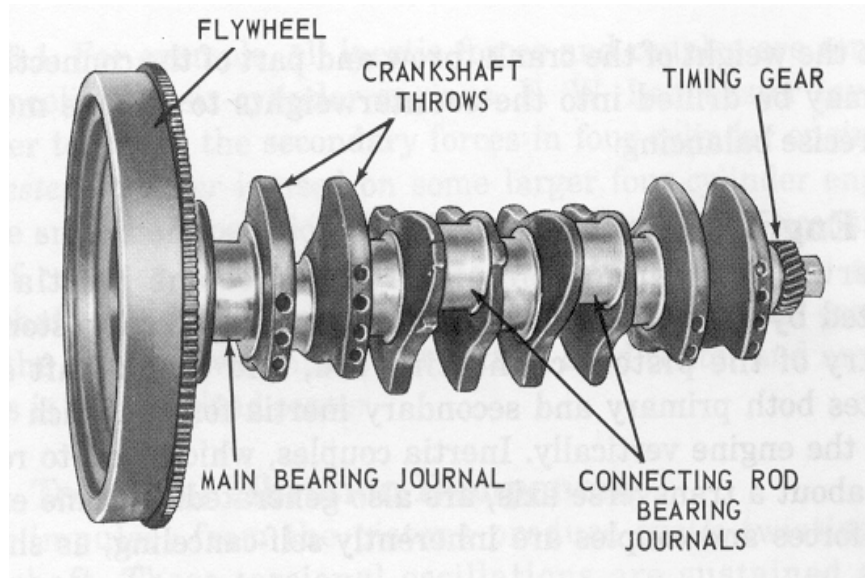
➤ Transmits mechanical energy from the engine to drives camshafts, generator, pumps, etc.

➤ Made of heat-treated steel alloys.

Moving Components

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❖ Flywheel

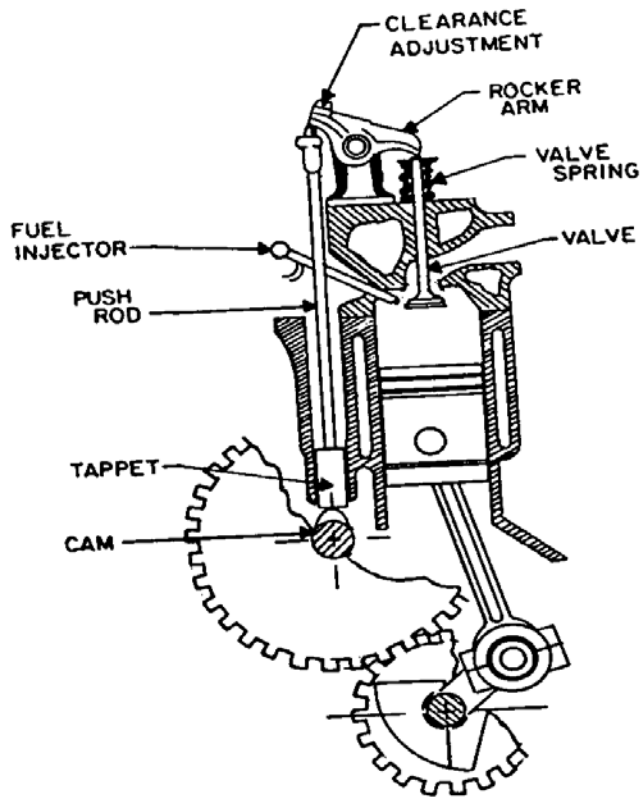


- Rotating mass with a large moment of inertia connected to the crankshaft of the engine.
- The purpose of the flywheel is to store energy and furnish a large angular momentum that keeps the engine rotating between power strokes and smoothes out engine operation.

Moving Components

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❖ Valve Train



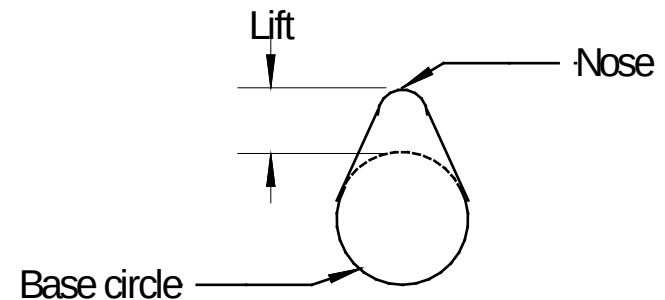
- Controls flow *into* and *out* of the combustion chamber.
 - Time and Duration
- Components (for OHV)
 - Camshaft
 - Valve tappets
 - Push rods
 - Rocker arm
 - Valves
 - Valve springs
 - Valve rotators
 - Valve seats

Moving Components

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❖ Camshaft & Cams

- Used to time the addition of intake and exhaust valves
- Operates valves via pushrods & rocker arms
- Driven by gear (or chain) from the crankshaft.
- 2:1 crankshaft to camshaft gear ratio.

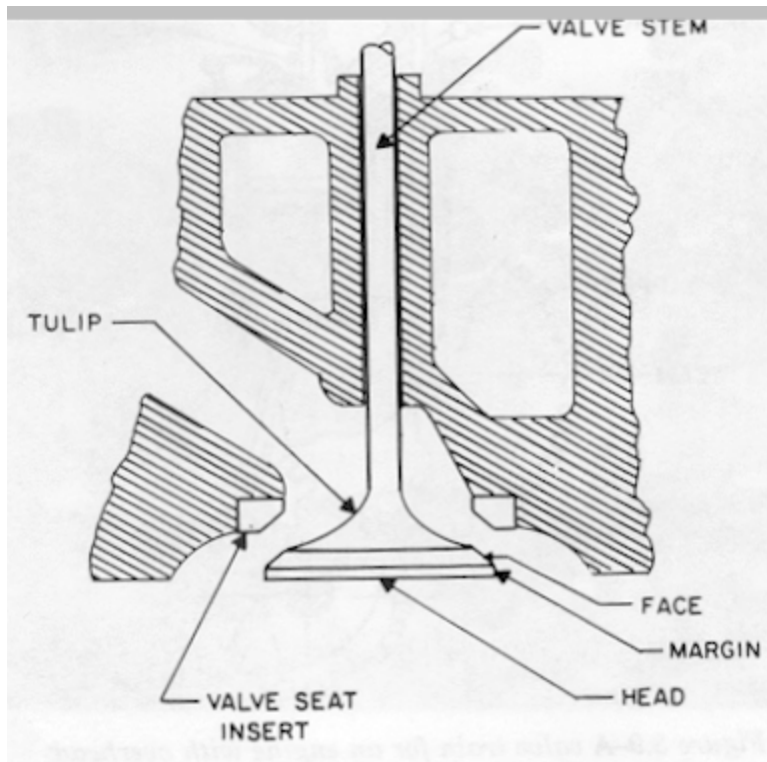


Cam Profile

Moving Components

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❖ Valves



- ▶ Each cylinder will have:
 - Intake: open to admit air to cylinder (with fuel in Otto cycle)
 - Exhaust: open to allow gases to be rejected

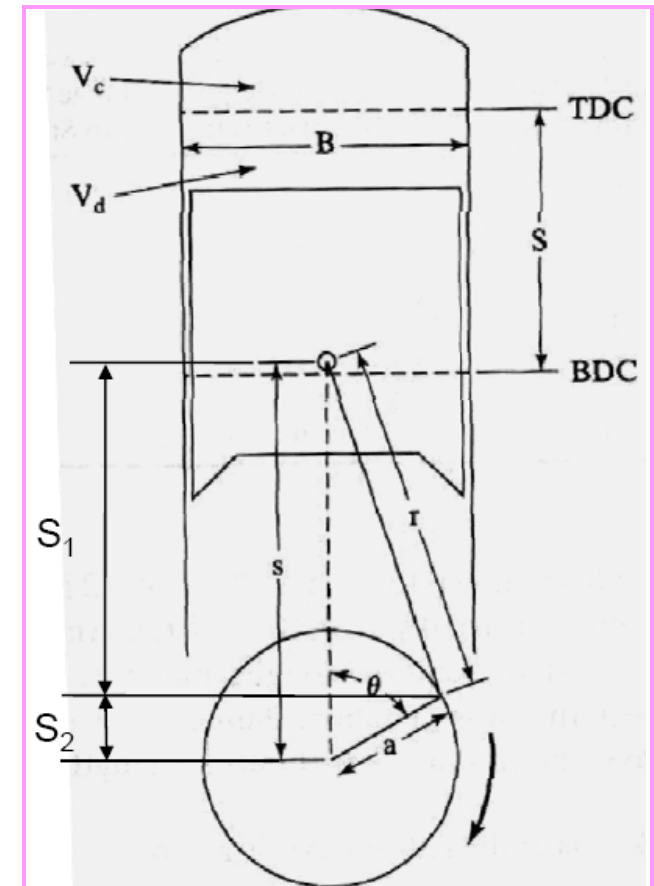
- ▶ Valve nomenclature
 - Head
 - Margin
 - Face
 - Tulip
 - Stem

ENGINE NOMENCLATURE

Engine Nomenclature

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- Cylinder Bore (B)
 - The inside diameter of the cylinder, and is measured in mm.
- Piston Area (A)
 - The area of circle diameter equal to the cylinder bore
- Stroke (S)
 - The linear distance, measured parallel to the axis of the cylinder, between the extreme upper and lower positions of the piston, measured in mm.

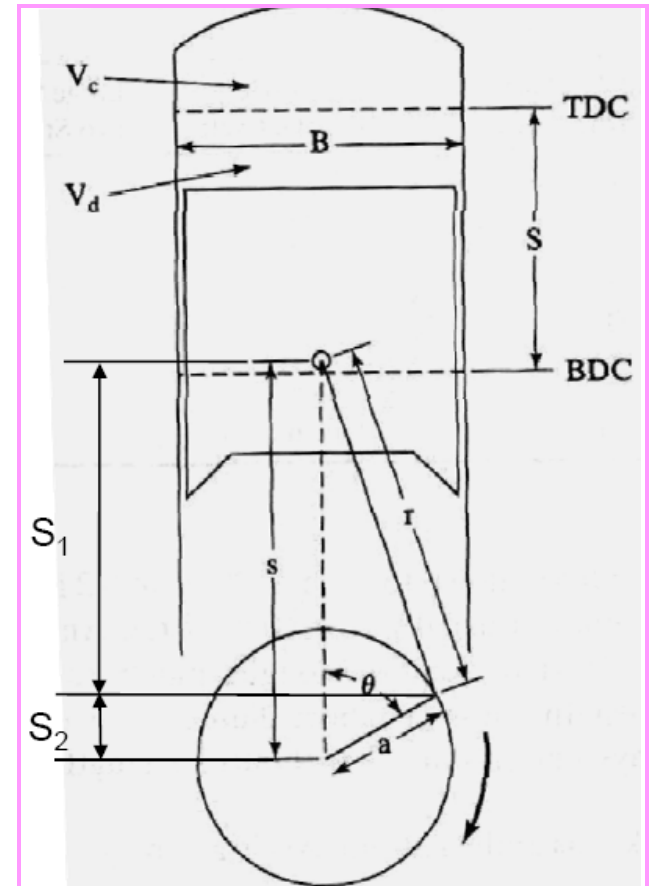


Engine Nomenclature

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Dead Centers

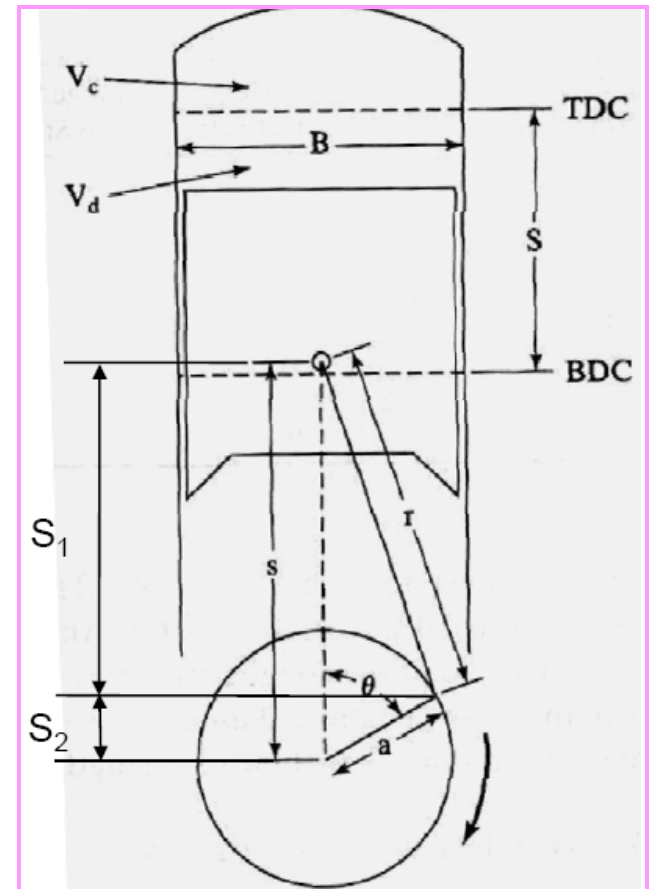
- The position of the working piston at the moment when the direction of piston motion reversed at either end of the stroke
- Top Dead Center (TDC) or Inner Dead Center (IDC) :-
 - when the piston is at farthest distance from the crankshaft



Engine Nomenclature

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- ⦿ Bottom Dead Center (BDC) or Outer Dead Center (ODC):- when the piston is nearest to the crankshaft
- Displacement Volume (V_d)
 - ▣ The nominal volume swept by the working piston when traveling from one dead center to the other
 - $V_d = A \times L = \pi/4(B^2L)$



Engine Nomenclature

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- Clearance Volume (V_c)
 - ▣ The nominal volume of the combustion chamber above the piston when it is at TDC is the clearance volume.
- Compression Ratio (r)
 - ▣ It is the ratio of the total cylinder volume when the piston is at the BDC, V_T , to the clearance volume V_c

$$r = \frac{V_T}{V_c} = \left(\frac{V_c + V_s}{V_c} \right) = 1 + \left(\frac{V_s}{V_c} \right)$$

